Warm-Up Answer Key

Integers and Coordinate Plane

LESSON 7-1 · EXPLORE INTEGERS AND THEIR OPPOSITES

MOVE *Name the opposite of a number and say which side of 0 it sits on*
SLIDE 1) 8 2) −1, −2 3) 6 4) 8x
MODEL 6 right 6

1. 9	**5.** to the left
2. −4	**6.** 0
3. 1/2	**7.** 15
4. −1, −2, −3	**8.** −12; the opposite is 12, which means 12 feet above sea level

DROPPED *skill 4 – simplifying an algebraic expression is carried over from Unit 6 and is not used anywhere in this lesson's move; it stays on the slide as a diagnostic box but has no place on the worksheet*

LESSON 7-2 · REPRESENT RATIONAL NUMBERS AND THEIR OPPOSITES ON THE NUMBER LINE

MOVE *Name the two integers a fraction or decimal falls between*
SLIDE 1) 3 2) 2 and 3 3) 5.5 4) 1/4
MODEL 5.5 5 5.5

1. 2 and 3	**5.** −4.1
2. −3 and −2	**6.** −1/5 (−0.2)
3. 3.5	**7.** −7.8
4. −3 and −2	**8.** the opposite is 5/2; each point is 2.5 units from 0

LESSON 7-3 · UNDERSTAND ABSOLUTE VALUE OF RATIONAL NUMBERS

MOVE *Report distance on a number line as a positive number*
SLIDE 1) 4 units 2) −9 3) 7 4) 7 units
MODEL 4 3 7

1. 9	**5.** 7 units
2. 6	**6.** 5 units
3. 12 units	**7.** 10 and −10
4. 3/4	**8.** 60 feet from sea level; 85 feet apart

LESSON 7-4 · COMPARE AND ORDER INTEGERS AND RATIONAL NUMBERS

MOVE *Compare numbers by their place on a number line: left is less*
SLIDE 1) > 2) −6 3) > 4) <
MODEL −0.5 1.5 1.5

1. <	**5.** −8, −5, 0, 2
2. <	**6.** −2, −1.5, −1/2, 1
3. <	**7.** −5.5 (any number between them, such as −11/2)
4. <	**8.** Tuesday (−9), Monday (−4), Wednesday (2)

LESSON 7-5 · REPRESENT RATIONAL NUMBERS ON THE COORDINATE PLANE

MOVE *Use the signs in an ordered pair to move from the origin and name the quadrant*
SLIDE 1) 3 up 2) Quadrant II 3) 2 down 4) 6
MODEL right down IV

1. Quadrant IV

2. Quadrant III

3. 2 units left, 7 units up

4. (−6, 4)

5. Quadrant II

6. no; it is on the y-axis, 5 units below the origin

7. (−3, −8)

8. (2, −5), Quadrant IV

DROPPED *skill 4 — reading a scaled axis needs a pictured grid, and this worksheet carries no figures; it stays as a slide box, where it can be stated in words*

LESSON 7-6 · DETERMINE DISTANCE ON THE COORDINATE PLANE

MOVE *Find the distance between two points that share a coordinate*
SLIDE 1) −8 2) 8 3) (7, −2) 4) 8 units
MODEL −8 8 8

1. 8 units

2. 8 units

3. 9

4. 7 units

5. 3 units

6. 4 units

7. (7, 7)

8. 10 blocks

DROPPED *skill 3 — reading coordinates off a plotted point needs a pictured grid and is Lesson 7-5's own work; the worksheet hands students the coordinates in writing instead*

LESSON 7-7 · REPRESENT POLYGONS ON THE COORDINATE PLANE

MOVE *Find a side length from two vertices, then add the sides for perimeter / Find the area from those side lengths*
SLIDE 1) rectangle 2) 4 units 3) 14 units 4) 12 square units
MODEL 3 14 12

1. bottom 5 units, left 3 units

2. 16 units; 15 square units

3. 14 units

4. 20 square units

5. 24 units; 36 square units

6. 12 square units

7. 5 units

8. perimeter 22 meters; area 28 square meters